

feature

The first mammal in space, the dog Laika, never returned safely to earth



A SPACE ODYSSEY

We take a look at current and future plans to conquer the final frontier

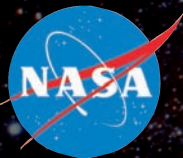
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If there was just one good thing that the world was managed to do, it would have to be the kick-start they gave to space activity. In 1942, at the peak of World War II, German scientists experimenting with extending missile ranges, sent the first rockets into space. These scientists were captured by both the Soviets and the United States after the war, and the same technologies used at war were later used for developing space capabilities. The Americans launched fruit flies into space in 1946, barely a year after the war, using captured

German rockets. This, however, was not seen as a threat by the Soviets at that time – and was not a part of the infamous space race. An integral part of the Cold War between the Soviets and the United States was the space race – similar to the arms race in many aspects.

The Russians heralded mankind's entry into space with the launch of the Sputnik I in 1957, which is significant for triggering the space race, and, as a result, the space age. The US didn't want to be too far behind and launched its own Explorer 1 in early 1958. America was under tremendous pressure to launch a space craft, as the Soviets stood to gain considerable tactical

A roundup of major space agencies from around the world



The National Aeronautics and Space Administration (NASA) is the official space agency for the United States. It is one of two space agencies most actively involved in the ISS project, and the NASA space shuttles bear half the burden of resupplying and re-manning the ISS. NASA has sent its astronauts further than anyone else, being the only space agency to have sent men to the moon. The Apollo missions and the shuttle missions are among the most significant achievements of NASA.



Roskosmos is the national space agency of Russia. The Soviet Space Program was at the forefront of the space race until the Apollo missions, being the first space agency to have sent a man into space. Yuri Gagarin orbited the earth aboard the Vostok 1 on 12 April 1961. The Soyuz spacecraft regularly resupplies the ISS. The RKA has sent the most number of space tourists aboard the ISS, through the agency of its affiliate, Space Adventures, Ltd. It is at the forefront of space habitat engineering, with the primary ISS modules being designed and fabricated in Russia.



The European Space Agency is primarily known for its launches of satellites. Most of the European Union is either actively involved, or have plans for future co-operation in this agency. The ESA has facilities in the UK, France, Italy, Germany, Spain, Netherlands and New Guinea. The ESA co-operates actively with most other space agencies, including the Chinese, the Japanese and the Indian space agencies. Current plans include building a Soyuz-like spacecraft for space travel beyond low-earth orbit (dubbed LEO in space jargon).



The Japanese Aerospace Exploration Agency is at the forefront of astronomy. It is responsible for a good portion of the tech wizardry in the ISS, most notably the Kibo lab. It is also planning manned missions to Mars; more seriously than anyone else. Jaxa collaborates heavily with the ESE. The Japanese are at the forefront of exotic space propulsion technology, especially in the domain of building and using solar sails.



The Chinese National Space Agency started in 1992, making it one of the youngest on the playground. However, it has shown stunning achievements in a short span of time, most notably with the Shenzhou program, putting a Chinese in orbit in 2003 – barely ten years after the agency's inception. The CNSA is one of the most secretive space agencies around, and likes to thrill the global community by revealing spectacular achievements suddenly and without pre-warning.



The Indian Space Research Organisation is on a par with the ESA in terms of satellite launches. ISRO has one of the largest fleet of launch vehicles, used for different kinds of launches, from polar to geosynchronous. India tested a spacecraft called the Space Capsule Recovery Experiment, which successfully remained in orbit for 12 days before returning to the Bay of Bengal. This proved that India was capable of manned spaceflight, which was more of an achievement than the widely publicised Chandrayaan mission. If you want to be an Indian astronaut (called a Brahmanaut), you can sign up from 2012 at Bangalore.